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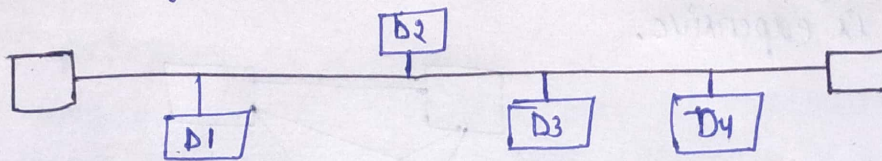
Subject - Computer Network

Experiment - To study of LAN using different Network topologies and different network connecting devices like Hub, switch, Repeater, bridge, and Router.

There are Five main types of network topologies used in computer network. Each topology describes how devices are connected and how data is flow in the network.

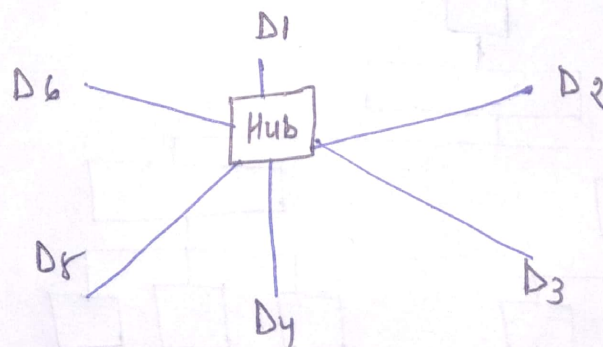
1) Bus Topology -

In Bus Topology, all devices are connected to a single main cable called the backbone. Data travels along this cable, and every device receives the signal but only the intended device accepts it. It is simple and inexpensive to install. However, if the main cable fails, the entire network stops working. It is suitable for small network.



2) Star Topology -

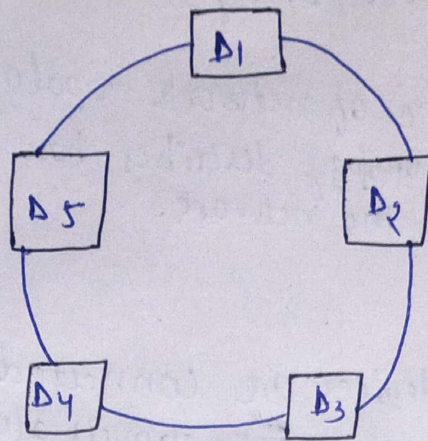
In star topology, all devices are connected to a central device such as a hub or switch. All communication passes through the central device. It is easy to manage and expand. If one cable fails, only that device is affected, but if the central device fails, the whole network stops.



3) Ring topology -

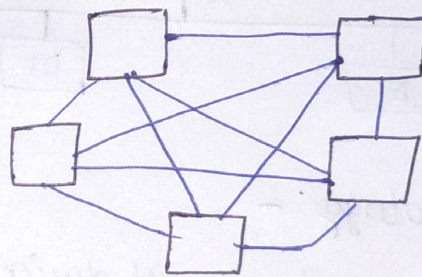
In ring topology devices are connected in a circular form. Each device is connected to two other devices. Data moves in one

one direction around the ring. If one device or connection fails, the entire network may be disrupted.



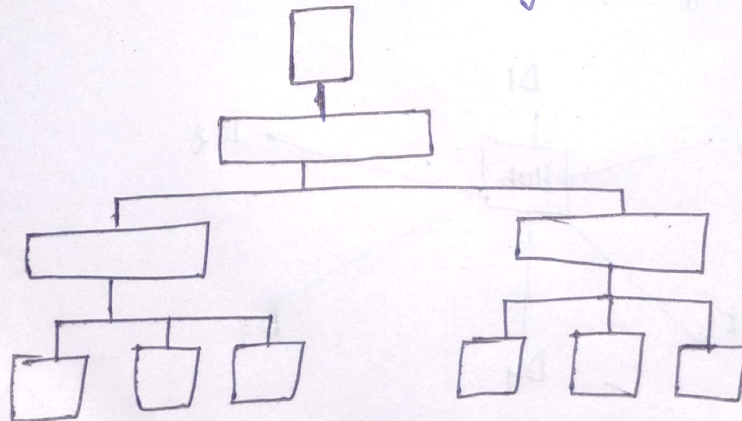
4) Mesh Topology -

In mesh topology, every device is connected to every other device. This provides high reliability because data can travel through multiple paths. However, it requires more cables and is expensive.



5) Tree Topology -

In tree topology, devices are arranged in a hierarchical structure like a tree. It ~~contains~~ combines star and bus topologies. It is suitable for large networks, but if the backbone cable fails, the whole network may be affected.



Network connecting devices -

Network connecting devices are hardware components used to connect computers and other devices in a network. They help in transmitting, receiving and managing data communication.

1) Hub -

A Hub is a basic networking device that connects multiple computers in a LAN. It sends (broadcasts) data to all connected devices. It works on the physical layer of the OSI model. It is slower and less effective bcz it does not filter data.

2) Switch -

A switch connects devices in a LAN and sends data only to the intended device using MAC addresses. It works on the Data Link layer. It is faster and more efficient than a hub and is commonly used in modern networks.

3) Repeater -

A Repeater is used to strengthen or regenerate weak signals so that data can travel longer distances. It works on the physical layer.

4) Bridge -

A Bridge connects two LAN segments and filters traffic using MAC addresses. It reduces network congestion and works on the Data Link layer.

5) Router -

A Router connects different networks together and allows communication b/w them. It connects LAN to the Internet and works on the network layer using IP addresses.